

TOGUCHIN, Russian Federation

WMO: 296360

Lat: 55.2246N

Lon: 84.3859E

Elev: 567

StdP: 14.40

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.5	-28.6	-39.3	0.6	-34.2	-33.4	0.8	-28.4	16.6	18.4	14.3	18.1	1.6	140	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	84.6	66.3	81.6	65.2	78.7	64.0	69.5	79.1	67.6	77.3	66.0	75.7	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.6	74.0	64.3	92.0	72.0	62.3	85.9	70.3	34.0	79.3	32.4	77.3	31.2	76.1	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.9	11.4	DB	-39.3	89.2	8.1	2.7	-45.1	91.1	-49.9	92.7	-54.5	94.2	-60.4	96.1	(4)
(5)			WB	-39.3	72.5	7.8	2.3	-44.9	74.2	-49.4	75.5	-53.8	76.8	-59.4	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	-0.7	5.9	21.5	38.2	51.7	62.3	65.8	60.9	49.5	36.7	18.6	5.6		
	DBStd	25.49	16.09	13.87	11.03	9.93	9.49	7.62	5.69	6.70	7.93	9.23	14.53	15.71		
	HDD50	7015	1572	1235	885	371	98	8	0	4	106	419	941	1376		
	HDD65	11202	2037	1655	1350	805	422	136	60	161	467	877	1391	1841		
	CDD50	1472	0	0	0	16	149	378	490	341	91	7	0	0		
	CDD65	186	0	0	0	0	9	55	85	34	3	0	0	0		
	CDH74	1947	0	0	0	5	196	606	732	359	49	0	0	0		
	CDH80	430	0	0	0	1	47	126	170	77	9	0	0	0		
	WSAvg	4.9	4.2	4.6	5.3	5.9	6.0	4.6	4.1	4.2	4.6	5.3	5.4	5.0		
	PrecAvg	19.0	0.9	0.7	0.8	1.1	1.6	2.2	2.8	2.4	1.9	1.7	1.5	1.3		
(15) Precipitation	PrecMax	23.2	1.7	1.8	1.7	1.9	3.2	3.9	5.1	4.1	3.8	3.5	3.0	2.3		
	PrecMin	13.3	0.1	0.1	0.1	0.3	0.4	0.7	0.4	0.6	0.6	0.8	0.7	0.4		
	PrecStd	2.7	0.4	0.4	0.4	0.5	0.8	0.9	1.1	0.9	0.8	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	35.7	48.2	71.8	86.2	87.3	88.7	87.1	80.4	66.3	48.3	35.9		
		MCWB	32.4	32.3	39.8	53.6	61.0	65.4	67.9	67.6	61.3	52.2	44.4	33.2		
	2%	DB	29.5	31.9	42.9	65.3	79.5	83.4	84.4	81.9	73.1	60.3	43.5	32.4		
		MCWB	27.4	29.4	36.6	51.2	59.2	64.8	68.1	66.1	58.5	49.2	39.7	30.8		
	5%	DB	25.9	28.9	39.4	59.8	74.9	80.5	81.1	78.0	68.2	55.4	39.1	29.6		
		MCWB	24.3	26.7	34.3	48.2	56.8	63.9	67.1	64.1	56.3	47.1	36.1	28.0		
	10%	DB	22.3	25.4	36.3	54.3	70.0	77.5	78.0	74.3	63.5	50.2	35.6	26.1		
		MCWB	20.7	23.5	32.5	45.2	54.7	63.0	65.8	62.7	53.8	44.5	33.2	24.5		
	0.4%	WB	32.4	33.1	40.5	55.9	63.9	69.3	72.6	70.5	63.0	53.8	44.7	33.7		
		MCDB	34.4	34.8	46.6	66.6	80.9	79.9	82.0	81.1	75.2	62.7	46.8	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.6	29.5	37.2	52.6	61.0	67.4	70.6	67.5	59.8	50.3	39.8	31.2		
		MCDB	29.1	31.7	41.6	64.1	75.9	79.0	80.0	77.6	71.0	58.5	42.8	32.5		
	5%	WB	24.2	26.8	34.9	49.0	58.4	65.9	68.9	65.7	57.3	47.6	36.4	27.9		
		MCDB	25.8	29.0	38.6	58.6	71.2	77.7	78.0	76.1	66.2	54.3	38.9	29.4		
	10%	WB	20.7	23.5	32.6	45.4	56.0	64.1	67.1	63.8	54.8	44.7	33.4	24.5		
		MCDB	22.2	25.3	36.0	53.7	68.0	74.4	75.6	72.6	61.9	50.3	35.4	26.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	17.3	18.0	18.2	21.0	19.9	18.4	19.4	18.6	14.2	12.0	13.8		
		MCDBR	13.8	14.2	18.9	27.9	29.7	25.8	23.7	25.8	27.7	24.3	12.4	13.7		
		MCWBR	12.9	12.8	13.3	15.9	13.4	10.6	9.8	11.5	13.9	14.6	9.6	13.2		
		MCDBR	13.6	14.6	16.7	25.8	26.0	22.0	20.1	22.6	24.8	22.1	12.1	13.5		
	5% WB	MCWBR	12.8	13.5	12.7	15.7	13.4	10.0	9.9	11.0	13.1	14.0	9.7	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.326	0.375	0.405	0.405	0.407	0.433	0.409	0.372	0.355	0.326	0.294		
	taud		2.006	1.961	1.902	2.058	2.203	2.284	2.209	2.264	2.313	2.221	2.037	1.952		
	Ebn at Noon		186	224	241	256	265	266	255	254	248	217	168	149		
	Edh at Noon		22	34	46	46	42	39	42	37	31	26	22	19		
(44) All-Sky Solar Radiation	RadAvg		288	613	1064	1463	1643	1894	1708	1407	948	477	241	181		
	RadStd		28	38	89	77	177	163	117	108	111	55	40	29		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page